Process Characterization Report

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCR-005

Process Development / Manufacturing Science & Technology (MSAT)

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1 Results

1.1 Response-surface models

The response-surface design was executed as 28 runs, made up of 16 factorial points, 8 axial points and 4 replicated centre points. A full quadratic model in the four factors was fitted to each response by ordinary least squares on the coded levels. Each model carries 14 terms beyond the intercept and leaves 13 residual degrees of freedom. The response-surface models are the predictive models for this step. The screening models in Table 1.5 and Table 1.6 identify which factors act and are not used to predict. Adequacy of the three fitted models is given in Table 1.1.

Table 1.1: Adequacy of the fitted response-surface models over the characterized region.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.989	0.976	0.932	80.9	2.71e-10	895
Step yield	28	0.913	0.818	0.586	9.69	0.000105	0.016
Leached Protein A (ppm)	28	0.567	0.101	-0.714	1.22	0.365	0.637

Pool host cell protein is described well across the whole characterized region. The model explains 0.989 of the variation in the response, at an adjusted R^2 of 0.976 and a predicted R^2 of 0.932, and it is significant at $p < 0.001$. The narrow gap between the adjusted and the predicted value means the model keeps its accuracy at settings where no run was made. Its residual standard error is 895 ng/mg against a set-point prediction of 7,216 ng/mg.

Step yield is described adequately. The adjusted R^2 is 0.818 and the predicted R^2 is 0.586, a wider gap than for host cell protein, so predictions of yield away from the design points carry more uncertainty. The residual standard error is 1.6% of product, against an observed spread of 84.4% to 99.0% across the study.

Leached Protein A is not described by the design at all. The model is not significant ($F = 1.22$, $p = 0.36$), the adjusted R^2 is 0.101, and the predicted R^2 is negative at -0.714. A negative predicted R^2 means the fitted surface predicts a new run less well than the response mean. Only 1 of the 14 terms reaches $\alpha = 0.05$, which is what chance alone returns at this number of terms. The replicated centre points give a %CV of 33.8 for leached Protein A, against 4.7 for pool host cell protein and 2.0 for step yield. The validated ELISA for leached Protein A (AMV-3016) has a precision of 6.5 %RSD and accounts for 41.0% of the variance of this response, so the spread at the centre point is not assay error alone. No coefficient of that model is carried into the interpretation below.

Table 1.2: Quadratic-model coefficients for pool host cell protein on coded factor levels. Factor codes: A protein load, B elution buffer ph, C load flow rate, D end of pool collect.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	7,215.5	309	23.3	5.44e-12	***
A	3,871.8	211	18.4	1.12e-10	***
B	-5,244.6	211	-24.9	2.4e-12	***
C	-435.4	211	-2.06	0.0596	
D	-1,055.5	211	-5	0.000242	***
AB	-1,899.8	224	-8.49	1.16e-06	***
AC	-429.1	224	-1.92	0.0774	
AD	-518.1	224	-2.32	0.0376	*
BC	348.2	224	1.56	0.144	
BD	727.4	224	3.25	0.00632	**
CD	-128.7	224	-0.575	0.575	
A^2	-344.5	557	-0.618	0.547	
B^2	1,943.5	557	3.49	0.00401	**
C^2	851.3	557	1.53	0.151	
D^2	330.5	557	0.593	0.563	

7 of the 14 terms in the host cell protein model are significant at $\alpha = 0.05$ (Table 1.2). Elution buffer pH carries the largest coefficient. Pool host cell protein falls by 5,245 ng/mg for each coded unit of pH, which is one half of the characterization range. Protein load is next in size and acts in the opposite direction, at 3,872 ng/mg per coded unit. The interaction between the two is significant at -1,900 ng/mg, and the quadratic term in pH is significant and positive, so the pH response is curved. End of pool collect enters as a smaller main effect, 5.0 times smaller than elution pH, together with its interactions with pH and with load. Load flow rate does not reach α on this response ($p = 0.060$).

Table 1.3: Quadratic-model coefficients for step yield on coded factor levels. Factor codes: A protein load, B elution buffer ph, C load flow rate, D end of pool collect.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.947	0.00554	171	3.5e-23	***
A	-0.0165	0.00377	-4.36	0.000767	***
B	-0.00243	0.00377	-0.643	0.531	
C	-0.00891	0.00377	-2.36	0.0346	*
D	0.0365	0.00377	9.68	2.62e-07	***
AB	0.00266	0.004	0.665	0.517	
AC	-0.00914	0.004	-2.28	0.0399	*

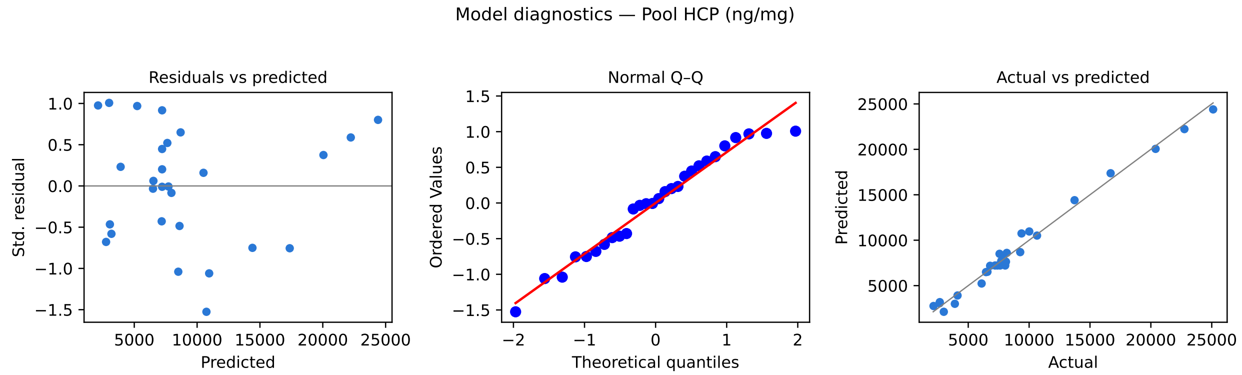
Term	Coef.	Std. err.	t	p-value	Sig.
AD	0.00629	0.004	1.57	0.14	
BC	0.00243	0.004	0.606	0.555	
BD	0.00291	0.004	0.727	0.48	
CD	0.00219	0.004	0.547	0.594	
A^2	-0.0128	0.00997	-1.29	0.221	
B^2	0.012	0.00997	1.2	0.252	
C^2	0.00681	0.00997	0.683	0.507	
D^2	-0.0178	0.00997	-1.78	0.0976	

4 terms are significant in the yield model (Table 1.3). End of pool collect dominates it, at 3.7% of product per coded unit. Protein load and load flow rate both reduce yield, and their interaction is significant. No quadratic term reaches α, the closest being the term in end of pool collect at $p = 0.098$, so the yield surface is close to planar over the region studied.

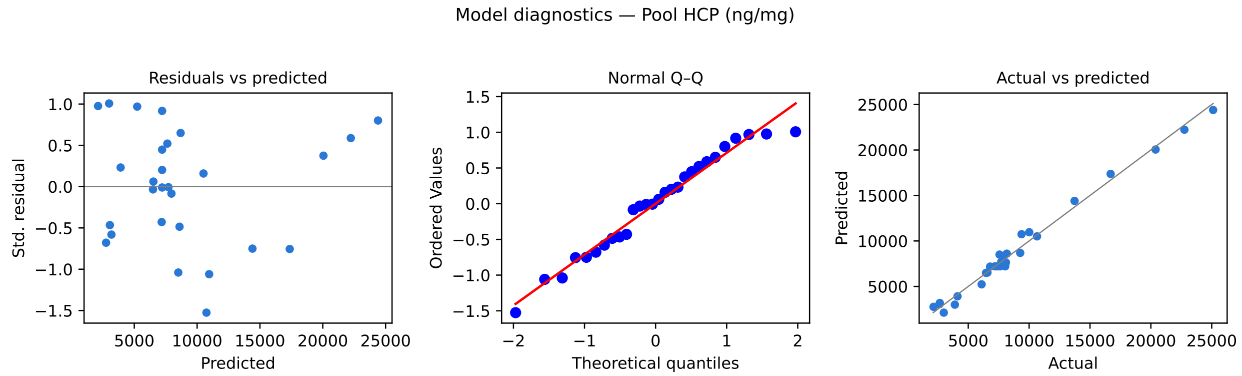
Table 1.4: Partition of the residual for pool host cell protein into lack of fit and pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.08e+08	14	6.48e+07	80.9	2.71e-10
Residual	1.04e+07	13	8.01e+05	nan	nan
Lack of fit	1e+07	10	1e+06	7.88	0.058
Pure error	3.82e+05	3	1.27e+05	nan	nan

Lack of fit for pool host cell protein does not reach $\alpha = 0.05$ (Table 1.4). The test returns $F = 7.88$ on 10 and 3 degrees of freedom at $p = 0.058$. Pure error rests on the 4 centre points alone, so the test has little power, and the result is read as no demonstrated lack of fit rather than as proof that the quadratic form is complete. The same test returns $p = 0.73$ for step yield and $p = 0.95$ for leached Protein A.



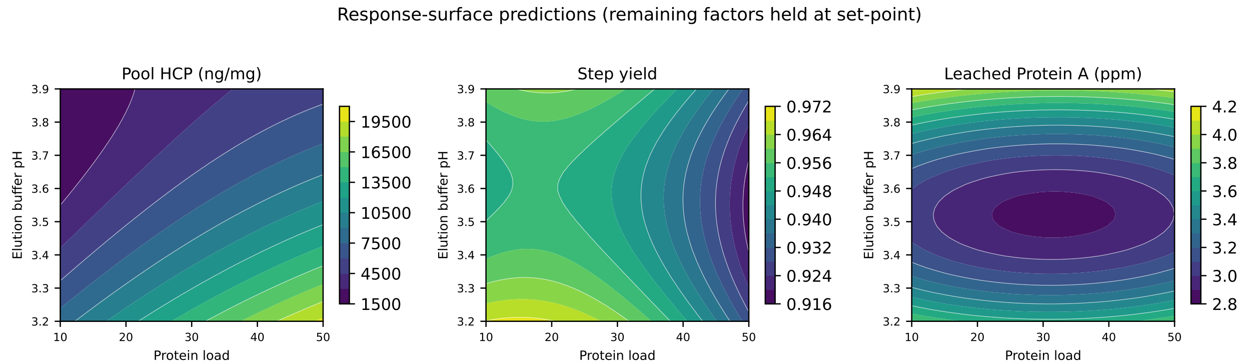
(a) Model diagnostics for pool host cell protein. Standardized residuals against predicted value, normal quantile plot of the standardized residuals, and actual against predicted.



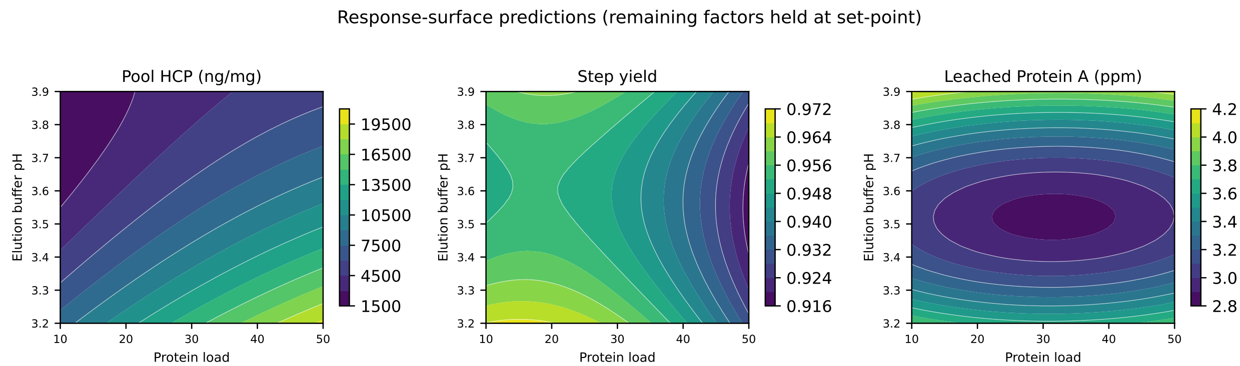
(b)

Figure 1.1

The diagnostics for the host cell protein model are shown in Figure 1.1. The residuals scatter without trend against the predicted value, and the largest standardized residual is 1.53. The normal quantile plot is close to linear and gives no evidence against the normality assumption. Actual and predicted values follow the identity line over the full range of the response. No run was excluded from any of the three fits.



(a) Fitted response surfaces over protein load and elution buffer pH, with the remaining two factors held at their set-points.



(b)

Figure 1.2

Figure 1.2 shows the three surfaces over protein load and elution buffer pH, with load flow rate and end of pool collect at their set-points. The host cell protein surface rises steeply toward the corner that combines high load with low elution pH, and it flattens along the load axis as the pH rises. The yield surface falls gently with load and depends little on elution pH over this range. The leached Protein A panel is shown for completeness and carries no structure the model supports.

1.2 Mechanistic interpretation

The two factors that dominate the host cell protein surface act on the same physical process. Host cell protein reaches the Protein A pool by two routes, adsorption to the base matrix and to the ligand, and association with the antibody already bound to the column. Elution buffer pH sets how much of that population leaves the column together with the product. A lower elution pH protonates the carboxyl and histidine groups at the contacts that hold these species and releases the weakly held host cell protein into the same volume as the antibody. The fitted model acts in that direction and gives elution pH the largest coefficient of any term in the study.

Protein load acts through the mass of antibody the bed carries. As the load approaches the dynamic binding capacity of the resin, the antibody occupies more of the accessible surface, the mass transfer zone extends toward the outlet, and more host cell protein is carried into the elution with the product. The main effect of load is positive and second in size only to elution pH.

The interaction between the two is the term that matters for operation. The load coefficient is not a constant, and differentiating the fitted model gives a slope of 5,772 ng/mg per coded unit of load at the bottom of the pH range and 1,972 ng/mg at the top. A high elution pH protects the pool against a high load. A low elution pH removes that protection. The two parameters cannot be ranged independently of each other.

The pH response is curved, and the direction of the curvature has a physical reading. The turning point of the fitted pH term lies at pH 4.02, above the top of the characterization range at pH 3.9. Pool host cell protein therefore improves as elution pH rises across the whole studied range, but by less at each step. That is the behaviour expected once the eluent is no longer acidic enough to desorb the weakly bound species, at which point a further rise in pH acts mainly on the product itself. Elution pH has no significant effect on step yield over this range, so the loss of product recovery that must eventually limit a higher elution pH lies above the range studied here.

End of pool collect lowers pool host cell protein and raises step yield at the same time, which is unusual for a purity and recovery pair. Pool host cell protein is expressed per milligram of product, and the antibody continues to elute after most of the co-eluting host cell protein has passed. Collecting further into the peak therefore adds product mass faster than it adds impurity, and the ratio falls. The interaction terms follow the same reading. Extending the collection lowers the pool by 1,783 ng/mg per coded unit at the bottom of the pH range and by only 328 ng/mg at the top, and the same gain is larger at high load than at low load. Those are the two conditions that put the most host cell protein into the front of the peak. The effect is monotonic across the characterized range and the model gives no sign of a rising impurity tail, but the design carries no information above 3.2 CV and no claim is made there.

Load flow rate acts on recovery rather than on selectivity. Binding of the antibody to the immobilized ligand is limited by diffusion into the resin pores, so a higher linear velocity shortens the residence time in the bed and broadens the breakthrough front during load. The fitted yield model shows the coupling this predicts. The flow coefficient is 0.02 percentage points of yield per coded unit at the bottom of the load range and -1.80 percentage points at the top. At a low load the breakthrough front stays inside the bed whatever the velocity, and product is lost only when a short residence time is combined with a load near the capacity of the resin. Load flow rate does not reach α on pool host cell protein.

Leached Protein A did not respond to any of the four parameters. Acid hydrolysis of the immobilized ligand depends on pH and would be expected to rise at the bottom of the elution range, and no such term was resolved above the variation of the assay. The level itself is the reason this matters little. The highest of the 47 characterization pools assayed 4.61 ppm against the drug substance limit of 5 ppm, and 0 of them exceeded it. That margin of 0.39 ppm is not large next to the precision of the method, so the attribute is not left to the elution conditions. Control rests on resin life cycle and sanitization under SOP-2008 and on the clearance credited to the polishing steps.

Three consequences follow for the way the step is operated. The first is that the elution buffer pH is the control point for host cell protein clearance at this step. 9 of the 47 characterization pools exceeded the in-process limit of 10,724 ng/mg, and 9 of those were eluted at the bottom of the characterization range, pH 3.2. The second is that load and pH have to be constrained together. At the top of the load normal operating range, 40 g/L resin, the model holds the pool at the in-process limit only above pH 3.46, which sits inside the normal operating range for pH but above its lower edge. On a grid across the four normal operating ranges, 93.8% of the settings meet the in-process limit for that reason, and the worst of them combines protein load 40 g/L resin, elution buffer pH 3.4, load flow rate 150 cm/hr and end of pool collect 2.3 CV, where the model predicts 13,553 ng/mg. The third is that the consequence of reaching that corner is bounded. Cation and anion exchange together clear host cell protein 483-fold, which leaves 28 ng/mg at the drug substance against a limit of 100 ng/mg. The step is therefore operated with elution pH held at or above its set-point when the load is at the top of its range, and the joint constraint is carried into the design space below rather than expressed as four independent ranges.

Appendix — screening effect tables referenced in the text

Table 1.5: Screening effect estimates for pool host cell protein, the six largest by magnitude.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-8,840.3	-4,420.2	350	-12.6	1.45e-06	***
A	7,269.4	3,634.7	350	10.4	6.4e-06	***
AB	-2,586.7	-1,293.4	350	-3.7	0.00608	**
AD	1,132.9	566.4	350	1.62	0.144	
AC	830.1	415.1	350	1.19	0.27	
D	707.9	353.9	350	1.01	0.341	

Table 1.6: Screening effect estimates for step yield, the six largest by magnitude.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	0.0604	0.0302	0.00208	14.5	5.01e-07	***
A	-0.0255	-0.0128	0.00208	-6.13	0.00028	***
C	-0.0237	-0.0118	0.00208	-5.68	0.000463	***
AC	-0.0209	-0.0104	0.00208	-5.01	0.00104	**
CD	-0.0107	-0.00537	0.00208	-2.58	0.0326	*
BD	-0.00981	-0.0049	0.00208	-2.36	0.0463	*